

YIFAN JIANG

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EDUCATION

The University of Texas at Austin, Austin, USA

2020 – 2024

Ph.D. in Electrical and Computer Engineering¹

- Advisor: [Zhangyang \(Atlas\) Wang](#)
- Committee: [Alan Bovik](#), [Sandeep Chinchali](#), [Mingyuan Zhou](#), [David Z. Pan](#)

Research Interests: Generative Model, Neural Rendering, Computational Photography

Huazhong University of Science and Technology, Wuhan, China

2015 – 2019

B.E. in Electronic Information Engineering

PUBLICATIONS

(* indicates equal contribution)

- [ICLR] **Y. Jiang**, H. Tang, J. Chang, D. Xu, L. Song, Z. Wang and L. Cao, “Efficient-3DiM: Learning a Generalizable Single-image Novel-view Synthesizer in One Day”, *International Conference on Learning Representations*, 2024
- [CVPR] **Y. Jiang**, H. Peter, B. Mildenhall, D. Xu, J. T. Barron, and Z. Wang, “AlignNeRF: High-Fidelity Neural Radiance Fields via Alignment-Aware Training”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023
- [ECCV] **Y. Jiang***, D. Xu*, P. Wang, Z. Fan, H. Shi, and Z. Wang. “SinNeRF: Training Neural Radiance Fields on Complex Scenes from a Single Image”, *European Conference on Computer Vision*, 2022.
- [ECCV] **Y. Jiang***, Z. Fan*, P. Wang*, X. Gong, D. Xu, and Z. Wang. “Unified Implicit Neural Stylization”, *European Conference on Computer Vision*, 2022.
- [ECCV] **Y. Jiang**, B. Wronski, B. Mildenhall, J. T. Barron, Z. Wang, and T. Xue. “Fast and High-Quality Image Denoising via Malleable Convolutions”, *European Conference on Computer Vision*, 2022.
- [WACV] **Y. Jiang**, X. Gong, J. Wu, H. Shi, Z. Yan, and Z. Wang, “AutoX3D: Searching Ultra-Efficient Architecture for Video Understanding”, *IEEE Winter Conference on Applications of Computer Vision*, 2022.
- [NeurIPS] **Y. Jiang**, S. Chang, and Z. Wang, “TransGAN: Two Pure Transformers can Make One Strong GAN and That Can Scale Up”, *Advances in Neural Information Processing Systems*, 2021.
- [ICCV] **Y. Jiang**, H. Zhang, J. Zhang, Y. Wang, Z. Lin, K. Sunkavalli, S. Chen, S. Amirghodsi, S. Kong, and Z. Wang, “SSH: A Self-supervised Framework for Image Harmonization”, *IEEE International Conference on Computer Vision*, 2021.
- [TIP] **Y. Jiang**, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, “EnlightenGAN: Deep Light Enhancement without Paired Supervision”, *IEEE Transaction on Image Processing*
- [CVPR] V. Goel, E. Peruzzo, **Y. Jiang**, D. Xu, N. Sebe, T. Darrell, Z. Wang, and H. Shi, “PAIR-Diffusion: Object-Level Image Editing with Structure-and-Appearance Paired Diffusion Models”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [NeurIPS] Z. Wang, **Y. Jiang**, Y. Lu, Y. Shen, P. He, W. Chen, Z. Wang, M. Zhou, “In-Context Learning Unlocked for Diffusion Models”, *Advances in Neural Information Processing Systems*, 2023. **Spotlight**
- [NeurIPS] Z. Wang, **Y. Jiang**, H. Zheng, P. Wang, P. He, Z. Wang, W. Chen, M. Zhou, “Patch Diffusion: Faster and More Data-Efficient Training of Diffusion Models”, *Advances in Neural Information Processing Systems*, 2023.
- [TMLR] Q. Wu, X. Chen, **Y. Jiang**, Z. Wang, “Chasing Better Deep Image Priors Between Over-and Under-parameterization”, *Transactions on Machine Learning Research*, 2023
- [CVPR] D. Xu, **Y. Jiang**, P. Wang, Z. Fan, Y. Wang, and Z. Wang, “NeuralLift-360: Lifting An In-the-wild 2D Photo to A 3D Object with 360° Views”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. **Highlight (2.5% of 9155 submissions)**
- [ICLR] Z. Fan, P. Wang, **Y. Jiang**, X. Gong, D. Xu, and Z. Wang, “NeRF-SOS: Any-View Self-supervised Object Segmentation on Complex Real-World Scenes”, *International Conference on Learning Representation*, 2023

¹Studied at Texas A&M University from Aug. 2019 to Aug. 2020; then transferred with my advisor to UT Austin

- [[NeurIPS](#)] D. Xu*, P. Wang*, **Y. Jiang**, Z. Fan, and Z. Wang, “Signal Processing for Implicit Neural Representations”, *Advances in Neural Information Processing Systems*, 2022.
- [[MM](#)] D. Xu, H. Poghosyan, S. Navasardyan, **Y. Jiang**, H. Shi, and Z. Wang, “ReCoRo: Region-Controllable Robust Light Enhancement by User-Specified Imprecise Masks”, *ACM Multimedia*, 2022
- [[TIP](#)] Z. Chen, **Y. Jiang**, D. Liu, and Z. Wang, “CERL: A Unified Optimization Framework for Light Enhancement with Realistic Noise”, *IEEE Transaction on Image Processing*
- [[NeurIPS](#)] B. Pan, R. Panda, **Y. Jiang**, Z. Wang, R. Feris, and A. Oliva, “IA-RED2: Interpretability Aware Redundancy Reduction for Vision Transformers”, *Advances in Neural Information Processing Systems*, 2021.
- [[ICLR](#)] T. Meng*, X. Chen*, **Y. Jiang**, and Z. Wang, “A Design Space Study for LISTA and Beyond”, *International Conference on Learning Representations*, 2021.
- [[DAC](#)] Y. Fu, Z. Yu, Y. Zhang, **Y. Jiang**, C. Li, Y. Liang, M. Jiang, Z. Wang, and Y. Lin, “InstantNet: Automated Generation and Deployment of Instantaneously Switchable Precision Networks”, *Design Automation Conference*, 2021.
- [[ICCV](#)] X. Gong, S. Chang, **Y. Jiang**, and Z. Wang. “AutoGAN: Neural Architecture Search for Generative Adversarial Networks”, *IEEE International Conference on Computer Vision*, 2019.

EXPERIENCE

[Apple](#), San Jose, USA May. 2023 – Present

Research Intern with AI/ML Argos Team, worked with [Liangliang Cao](#), [Hao Tang](#), and [Rick Chang](#)

- Working on the 3D generative model.

[Adobe](#), San Jose, USA May. 2022 – Dec. 2022

Research Intern with [Marc Levoy's Team](#), worked with [Zhihao Xia](#), [Cecilia Zhang](#), [Xiuming Zhang](#) [Jiawen Chen](#).

- Developed a high-quality depth estimator using multi-sensor assistance. The resultant paper is under review.

[Google Research](#), Mountain View, USA May. 2021 – May. 2022

Research Intern with [GCam](#), worked with [Tianfan Xue](#), [Bart Wroński](#), [Ben Mildenhall](#), [Peter Hedman](#), [Jon Barron](#).

- Developed a fast denoising operator as well as an efficient backbone. The resultant paper was accepted by ECCV'2022
- Designed a high-fidelity neural radiance field that can render high-quality novel view images. The resultant paper was accepted by CVPR'2023

[Adobe](#), San Jose, USA May. 2020 – Nov. 2020

Research Intern with Applied Research Team (ART), worked with [He Zhang](#) and [Jianming Zhang](#).

- Developed a self-supervised method for image harmonization. The resultant paper was accepted by ICCV'2021

[Bytedance AI Lab](#), Beijing, China Jan. 2019 – Aug. 2019

Research Intern with US CV Lab, worked with [Xiaohui Shen](#), [Ding Liu](#), [Chen Fang](#), [Jianchao Yang](#).

- Designed a jointly image denoising and low-light enhancement algorithm for a selfie camera app [FaceU](#).

MEDIA HIGHLIGHT

- TransGAN was covered by [Quanta Magazine](#) (Mar. 2022) and was highlighted by [Top AI influencers](#) and high-profile [YouTubers](#), as well as considered as [the most influential new paper of the month \(Feb. 2021\)](#).
- AutoGAN was covered by [Synched AI Technology & Industry Review](#) (Aug. 2019), and also featured on [Towards Data Science](#) (Sep. 2019) and [Analytics Magazine](#) (Aug. 2019), etc.

COMMUNITY SERVICES

- Reviewer for: CVPR, ICCV, ECCV, ICML, NeurIPS, ICLR, Siggraph, Siggraph Aisa, IROS, WACV, IJCAI, TPAMI, TIP, IJCV, NeuroComputing, RA-L, TCI, TCSVT
- Workshop Organizer for: [ECCV RLQ-TOD Workshop 2020](#)

INVITED TALKS

- Invited Talk at Adobe.
- Invited Talk at TikTok.

- Invited Talk at Meta GenAI.
- Invited Talk at Google.
- “Improving Diffusion Model for Novel View Synthesis: Faster Training and Better Convergence”, at Apple Core Algorithm Org.
- “Extreme Novel-view Synthesis and 3D Content Creation using Deep Generative Prior”, at [Caltech & UCLA](#) and Apple Video Computer Vision team.
- “Learning to Enhance Low-light Images without Paired Supervision” at [\[IEEE SPS Webinar\]](#), after my TIP work EnlightenGAN was highlighted as one of SPS’s top-25 most downloaded articles on IEEE Xplore®, 2021-2022.
- “Fast and High-Quality Image Denoising via Malleable Convolutions” at Adobe, [Marc Levoy’s team](#).
- “Vision Transformer for Image Generation, Editing, and Processing” at Google Research, GCam.
- “TransGAN: Two Transformers Can Make One Strong GAN” at [\[cai-workshop\]](#), [\[SHI Lab @University of Oregon\]](#)

AWARDS

- [University of Texas Graduate Dean’s Prestigious Fellowship](#) 2023
- [Apple Scholars in AI/ML Fellowship](#) 2023

MENTORED STUDENTS

- Dejie Xu (M.S. student at UT-Austin) -> Now Ph.D. student at UT-Austin
- Zeyuan Chen (Undergrad at USTC) -> Now Ph.D. student at UCSD
- Qiming Wu (Undergrad at HUST) -> Now M.S. student at UCSB
- Qiucheng Wu (Ph.D. student at UCSB)